

ANSHUL RAMAN

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Education

University of Cincinnati — College of Engineering and Applied Science

B.S. Computer Science | GPA: 3.68 | Dean's List, CEAS

Cincinnati, OH

Expected May 2030

- **Relevant Coursework:** Computer Science I, Calculus I–II, Engineering Applications I–II
- **Activities:** IEEE, UC Robotics, AI Society, Bearcat Tech Innovators, Tennis Club

Experience

Founding Software Engineer, OpenPrem

April 2026 – Present

Cincinnati, OH

- Productized OpenPrem's AI infrastructure from prototype into Relix, reducing local setup from roughly 30 minutes across multiple services to **under 3 minutes** through single binary Windows deployment, 14 Rust crates, 90 API routes, SQLite state, a React dashboard, and 46 versioned releases.
- Built a governed AI agent runtime to make automation safer and more controllable, reducing unsafe tool execution risk by **28 percent** in internal policy tests through bounded reasoning loops, strict JSON decision contracts, approval gates, fail-closed validation, and self-correction.
- Developed SolFlow to replace code-heavy workflow setup with visual graph based creation, cutting sample workflow build time from 45 minutes to **under 10 minutes** while adding a Vue and TypeScript IDE, Rust SOL compiler, WebAssembly VM, real-time diagnostics, source tracing, and 407 automated tests.
- Took OpenPrem from engineering prototype to enterprise demonstration platform, expanding demo coverage from 1 prototype to a **15 deployment SaaS demo fleet** and 37 page management console with OIDC and SAML SSO, TOTP 2FA, RBAC, SSE monitoring, an LLM assistant, and a real-time mesh operations console.

Projects

ARIA — Ambient Reasoning & Intelligence Architecture

March 2026 – Present

Personal AI Operating System | Node.js, Next.js, FastAPI, Supabase, Qdrant, Voyage, Gemini, Playwright, ESP32

- Built ARIA, a personal AI operating system **used daily across real accounts**, to close the gap between chatbots and useful assistants by unifying **9 surfaces** including web chat, Telegram, Gmail, Calendar, browser history, AI conversations, ARIA OS, an execution agent, and ESP32 voice hardware under one shared brain.
- Engineered **persistent memory** so ARIA does not restart from zero every chat, indexing a 1,000 plus note GitHub vault and conversations from 6 AI platforms through Qdrant, Voyage, semantic search, reranking, and SHA diff indexing.
- Developed a **supervised execution agent** that turns natural language into verified action across browsing, code execution, research, task management, Telegram progress streaming, approval gates, fail-closed validation, and prompt injection isolation.
- Created ARIA OS as the secure control layer for the system, combining authenticated terminal and file access, task monitoring, self-hosted ATLAS OIDC and PKCE SSO, and an **ESP32 voice and vision node** with 1 to 2 second first audio latency.

VoxVitals — AI Clinical Trial Monitoring Platform

March 2026

Next.js, React, TypeScript, Supabase, Gemini, ElevenLabs, Presage SDK, YOLOv8

- Turned 10 minute manual symptom diaries into **under 90 second voice check-ins**, leading full stack architecture for patient, clinician, and lead doctor workflows in 24 hours to win 3rd Place in the Medpace Clinical Trial Challenge and Best Use of Presage API from MLH.
- Built a clinical triage engine that cut demo patient review time from **8 minutes to under 60 seconds**, converting spoken check-ins into transcripts, 0 to 100 risk scores, 7 signal dropout predictions, 6 detector alerts, cohort analytics, and medication safety checks across 26 API routes, 6 RLS protected tables, and 23 React components.

Gradintel

February 2026 – March 2026

Vanilla JavaScript, Supabase, Canvas LMS, Chart.js, PDF.js, Pollinations, Groq, Gemini, Claude, OpenAI

- Launched Gradintel to **45 plus college and high school students across Ohio and California**, replacing manual GPA spreadsheets with a cloud-synced academic planning engine that processed 450 plus GPA simulations and calculated exact exam scores needed for target grades in seconds.
- Built the academic intelligence layer behind Gradintel, improving student planning accuracy to **86 percent** in user-checked scenarios through Canvas import across 3 paths, a 5 provider AI advisor, syllabus and degree PDF parsing, What-If simulation, GPA Time Machine, analytics dashboards, multi-degree planning, and export to JSON, CSV, text, and ICS.

Technical Skills

Languages: Python, TypeScript, JavaScript, Rust, C++, SQL, HTML, CSS

Frameworks & Tools: React, Next.js, Vue 3, Node.js, Express, FastAPI, Axum (Tokio), Supabase, Docker, Vercel, Git, WebAssembly

AI / ML: LLM orchestration (Gemini, OpenAI, Anthropic), RAG pipelines, Qdrant, MCP, agentic systems, wasm-bindgen

Awards & Activities

Gold Medal — National-Level Science Olympiad (2023)

Gold Medal — National Abacus Competition (2023)

Top 150 National — All India Tennis Association (AITA, 2022)

Volunteer: Mary Magdalen House (clothing intake); Bearcat Recycling (event waste sorting); Epic House (art for elderly residents)

Available for Co-op: Fall 2026